NATIONAL ECONOMICS UNIVERSITY

Faculty of Data Science and Artificial Intelligence

DATA ANALYSIS WITH PYTHON / ASSIGNMENT 06

Book Rating Classification

Can user and book information help identify ratings of 8–10?

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Predict the rating class, one rating at a time

8–10

Class 1: high rating

Threshold selected by the group

1–7

Class 0: other ratings

Does not mean the reader dislikes it

0

Implicit feedback

Retained for activity summaries

- Prediction unit: one observed user–book rating with a score from 1 to 10.
- Inputs: user and book metadata plus statistics from other training ratings.
- Scope: binary classification; these metrics do not evaluate recommendation ranking.

Source: notebook intro; report §1

Three tables, linked through users and books

Table	Raw rows	Key / content
Books	271,360	ISBN; title, author, year, publisher
Users	278,858	User-ID; location, age
Ratings	1,149,780	User-ID + ISBN; score 0–10

Ratings join Users on User-ID | Ratings join Books on ISBN

- Left joins preserve ratings even when book metadata are unavailable.
- Many-to-one validation prevents row multiplication; ISBN stays a string.

Data: Möbius / Kaggle; notebook §1–2

Clean metadata while preserving the rating sample

- Trim ISBNs; remove one exact duplicate book row after normalization.
- Quarantine three malformed book rows; retain their associated ratings.
- Age rule: 5–100. Publication-year rule: 1450–2026; year 0 means unknown.
- Keep missingness flags; learn numeric medians inside the training pipeline.

433,671

Explicit ratings

Final classification dataset

30.11%

Age missing

Among classification ratings

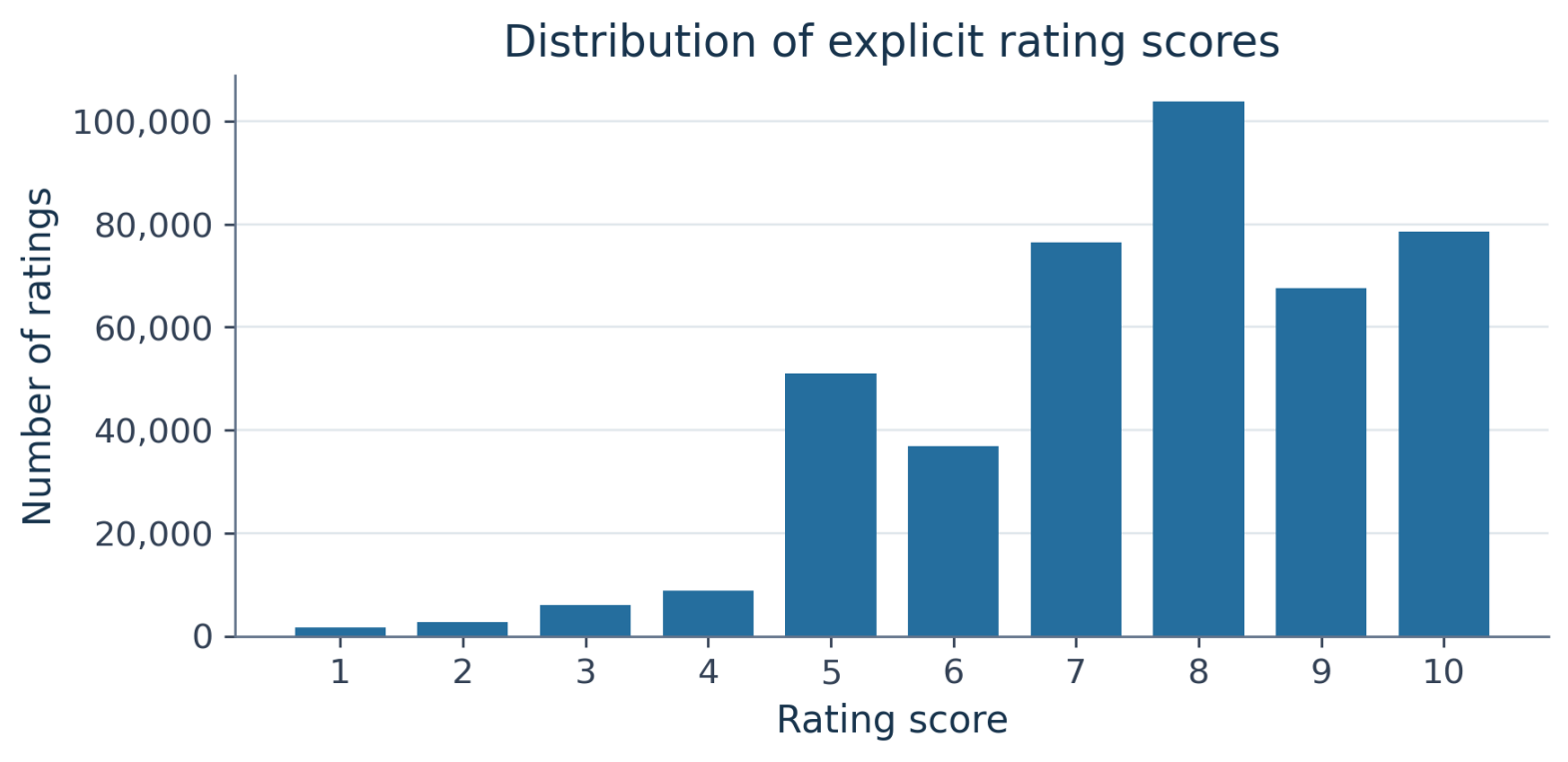
11.49%

Book metadata missing

49,829 classification ratings

Source: notebook §3; report §3

High ratings are common; a baseline is essential



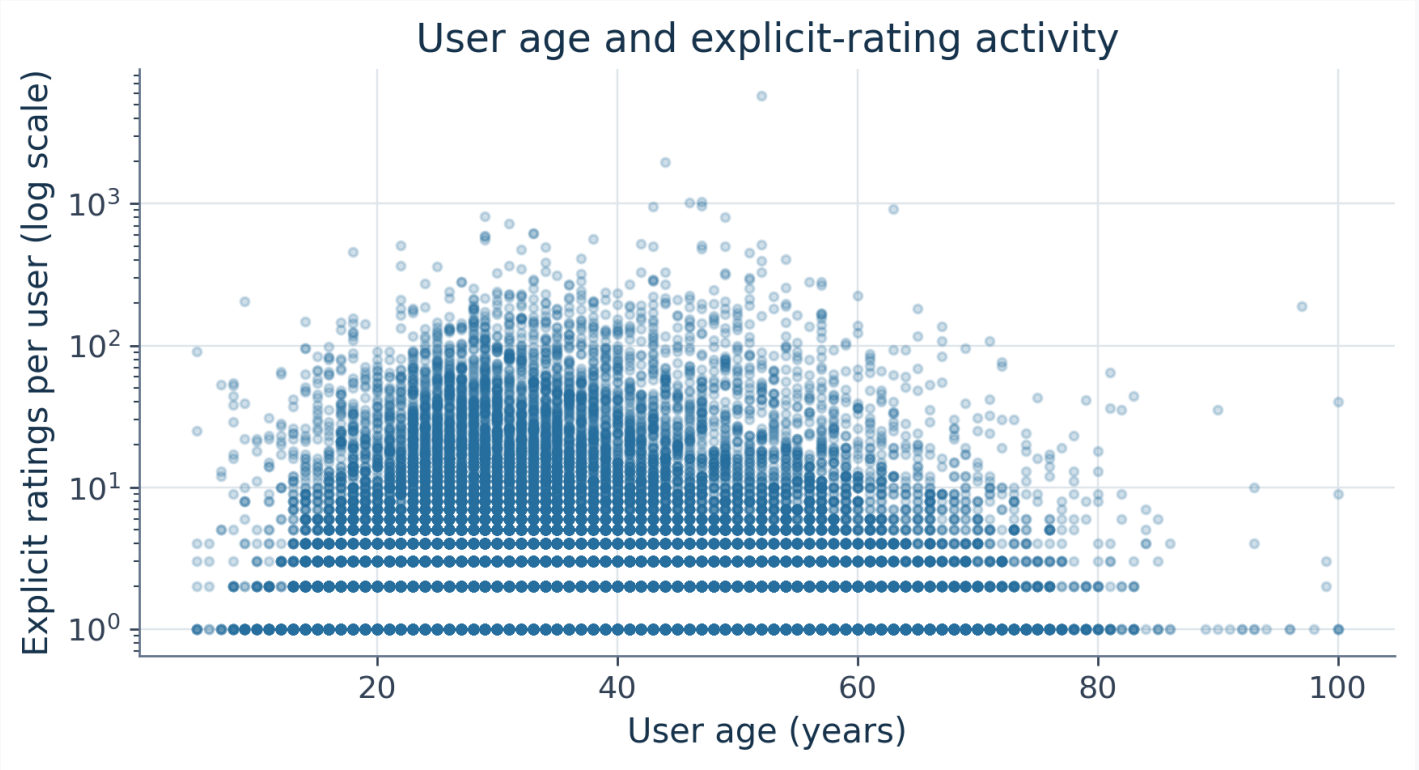
57.62%

of explicit ratings belong to class 1

- 249,887 high ratings; 183,784 ratings of 1–7.
- Score 8 is the most frequent explicit score.
- Always predicting class 1 already gives 57.62% Accuracy.

Source: notebook §4.2; report §4.1–4.2

Activity is uneven, and age alone is a weak signal

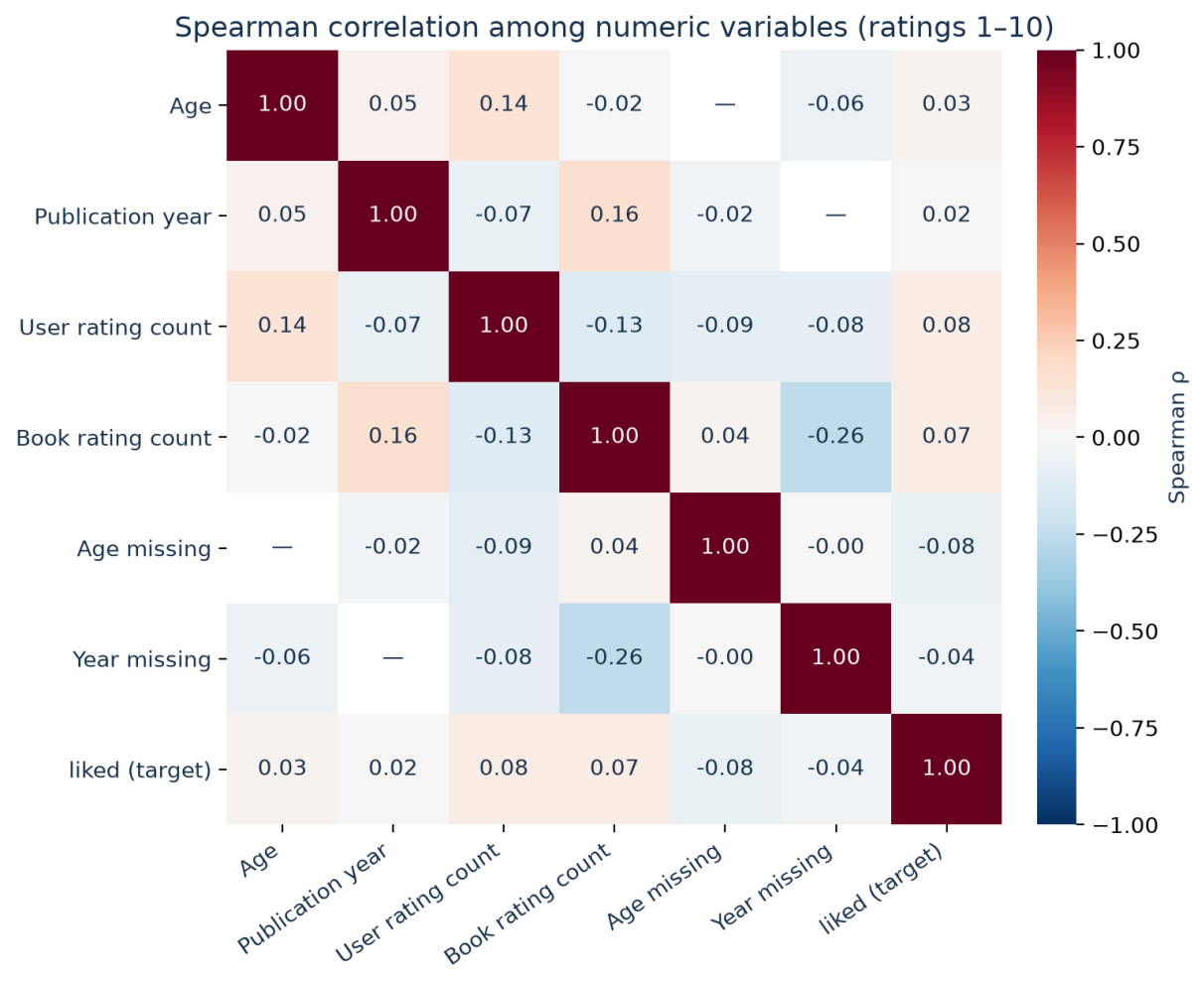


- Known user age: median 32; age is missing for 40.17% of all users.
- Age-group / rating-class association is small: Cramér's V is about 0.0483.
- User rating counts are highly uneven; test count and ID-rate features.

Repeated users and books violate simple independence assumptions; EDA is descriptive, not causal evidence.

Source: notebook §4.6–4.8; report §4.3–4.4

No single numeric variable separates the classes



- Spearman rho: rank-based, robust to the highly skewed rating counts.
- Every |rho| with **liked** is at most 0.08: Age missing -0.076, user count 0.075, book count 0.072.
- Strongest input pair: book count vs Year missing (-0.26); no serious multicollinearity.
- Weak one-to-one links do not rule out joint value; feature groups are chosen by validation.

Source: notebook §4.9; report §4.5

Separate fitting, selection and evaluation

70%

Train: 303,569

Fit preprocessing and estimators

15%

Validation: 65,051

Select one of 20 configurations

15%

Test: 65,051

Evaluate the selected pipeline

- Stratify by class; split seed 56. Same partitions for every configuration.
- Imputation, scaling and one-hot encoding learn only from train.
- Validation/test count and rate features look up train statistics only.
- Train target rates use 5-fold OOF encoding and smoothing (alpha = 20).

Source: notebook §5.2–5.5; report §5

Five cumulative feature groups

Group	New information	Inputs
Initial	Age, publication year, missingness flags	4
Metadata	Book author and publisher	6
Country	Country parsed from Location	7
Count	Training rating counts per user / book	9
Target Encoding	**OOF high-rating rates per user / book**	**11**

Book-Rating is excluded from X. User-ID and ISBN are lookup keys, not numeric predictors.

Unseen IDs: count = 0; target rate = training prior. OOF excludes each row’s own label.

Source: notebook §5.1–5.5; report §5.1, §5.3

20 configurations; one selection rule

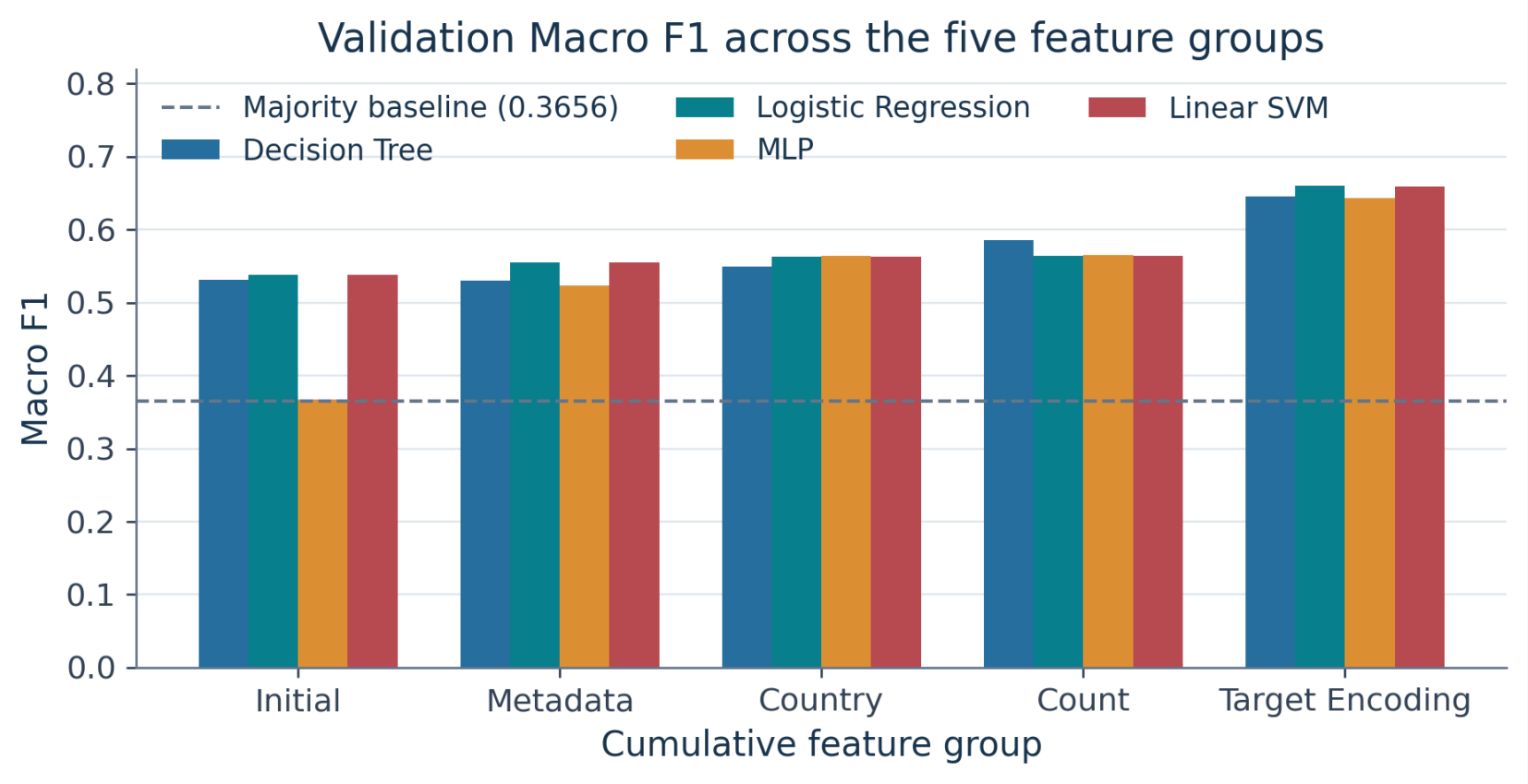
Algorithm	Fixed configuration across feature groups
Logistic Regression	Balanced class weights; max_iter = 1000
Decision Tree	Balanced; depth 20; minimum leaf size 50
Linear SVM	Balanced; LinearSVC; max_iter = 3000
MLP	Hidden layers (64, 32); training-loss stopping

Select the highest validation Macro F1

Tie-break: ROC-AUC, then Accuracy. Baseline: always predict the majority class.

Source: notebook §5.3–5.6; report §6

Target encoding gives the clearest gains



0.6597

Validation Macro F1
LR + Target Encoding

- LR Count: 0.5638
  LR Target Encoding: 0.6597
- SVM is close: 0.6594. A single split does not establish a stable advantage.

Selected across all 20 configurations using the same validation criterion.

Source: notebook §5.6; report §7

Better recognition of both classes, with trade-offs

Metric	LR + Target Encoding	Baseline
Accuracy	0.6565	0.5762
Macro F1	**0.6553**	**0.3656**
ROC-AUC	0.7306	0.5000
Class 0 Recall	0.7041	0.0000
Class 1 F1	0.6758	0.7311

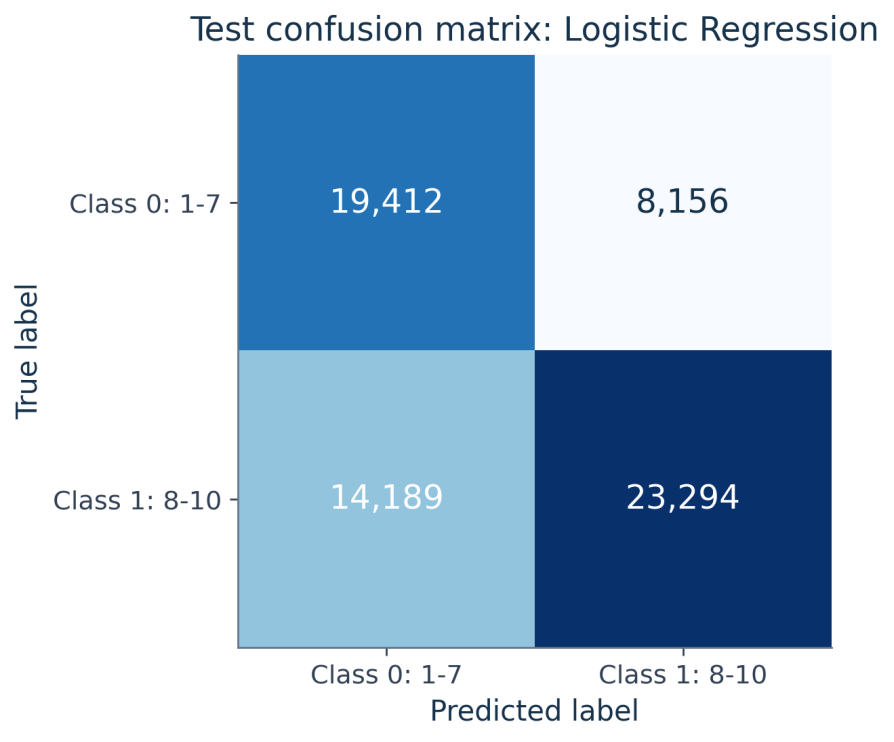
+8.03 pp Accuracy

+0.2897 Macro F1

Class 1 F1 falls versus the baseline. The selected model does not lead on every metric.

Source: notebook §5.7; report §8.1

Errors and discrimination on 65,051 test ratings





High-rating precision: 74.07%

23,294 correct out of 31,450 high predictions.

ROC-AUC: 0.7306

Discrimination across thresholds; not Accuracy or a calibration measure.

Source: notebook §5.7–5.8; report §8.2–8.3

Aggregate performance does not cover every ID group

Test segment	Ratings	Macro F1	Recall 1
Seen user, seen book	37,143	0.6589	0.6609
New user, seen book	5,513	0.5673	0.4963
Seen user, new book	19,908	0.6832	0.6361
New user, new book	**2,487**	**0.4234**	**0.1097**

- Both IDs new: rate features fall back to the prior; count features equal zero.
- Segments have different class mix and sample sizes. Use an ID-based split for stronger evidence.

Source: notebook §5.7; report §8.4

A useful signal, with a clear next validation step

WHAT WE LEARNED

- Training rating tendencies improve this classification experiment.
- LR + Target Encoding leads validation; test Macro F1 is 0.6553.

WHAT REMAINS LIMITED

- Random rating split shares IDs; missing metadata and OOF shift remain.
- EDA used the full sample; test was already viewed during development.

Next: a fresh holdout, user/book or time-based evaluation, repeated splits, and ranking metrics before recommendation use.

Materials: notebook + analysis report + this 15-slide presentation

github.com/Bluenip/Data_analyst_python_midterm_project

Source: notebook §6; report §9–10